

AI/BigData 인텐시브 과정

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AI/BigData 인텐시브 과정

Section 1. 환경 설정

Section 1-1. 가상 환경

가상 환경(virtual environment)

Anaconda

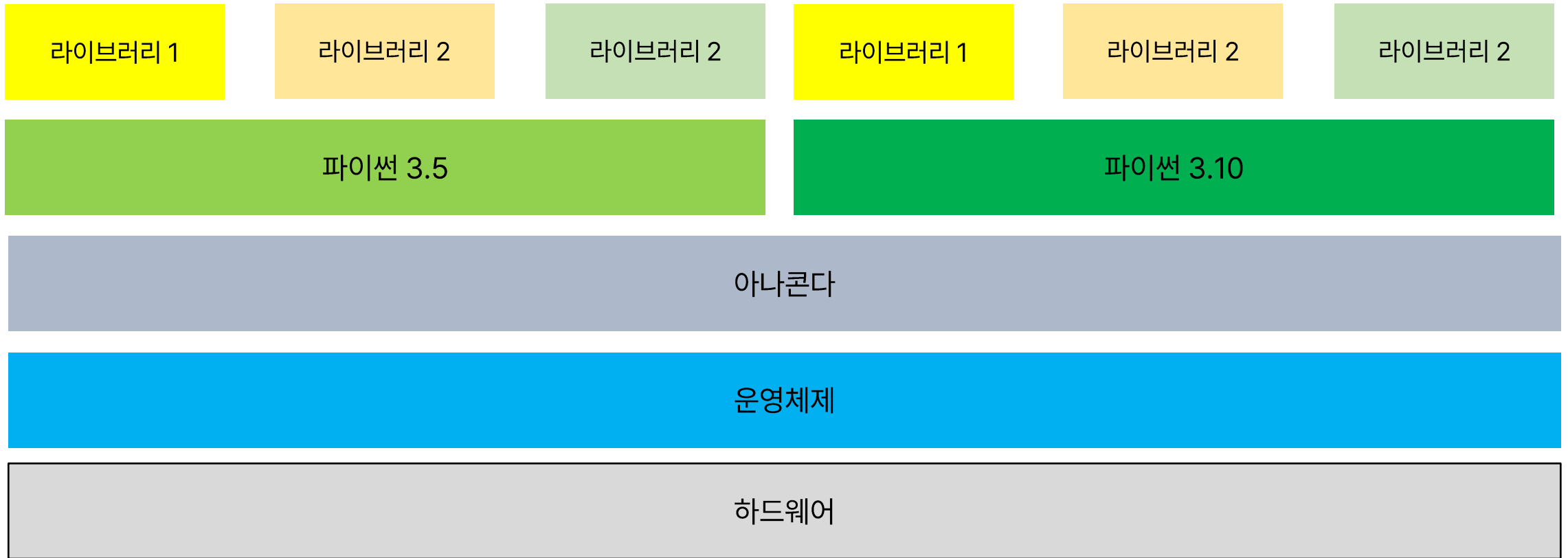
Python

Library

가상 환경(virtual environment)



가상 환경(virtual environment)

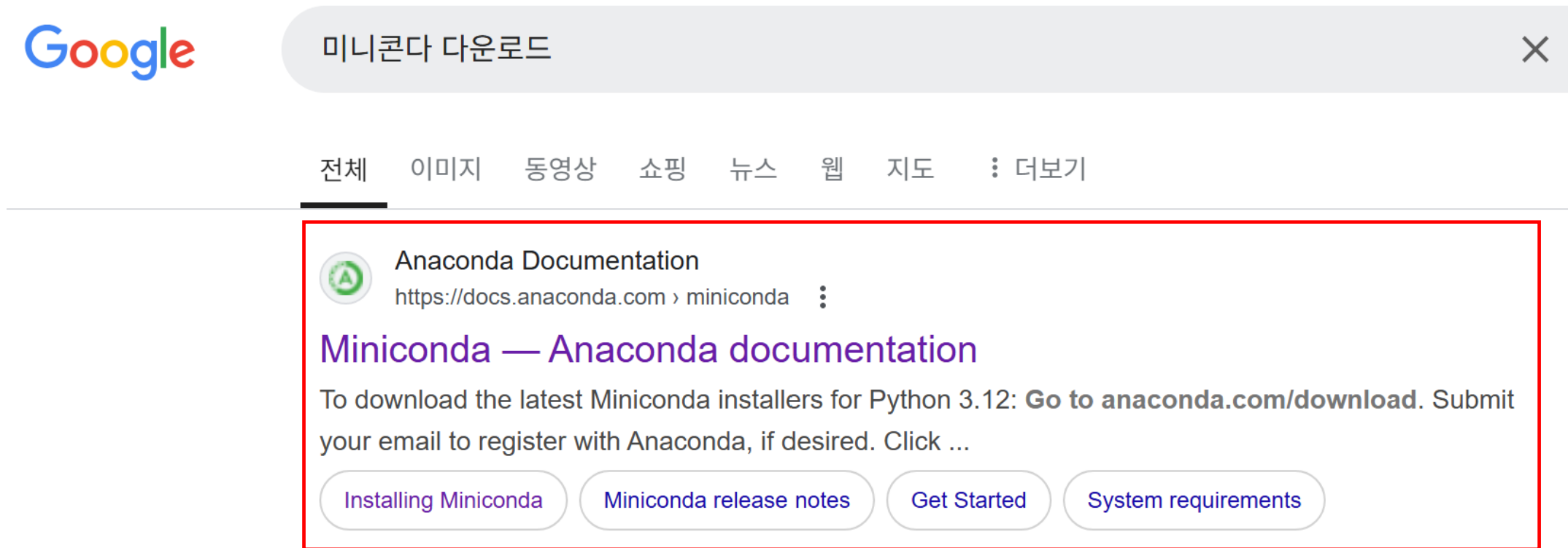


AI/BigData 인텐시브 과정

Section 1. 환경 설정

Section 1-2. 파이썬 설치

파이썬 설치 – 구글에서 '미니콘다 다운로드' 검색



파이썬 설치 - 구글에서 '미니콘다 다운로드' 검색

The screenshot shows the Anaconda documentation website for Miniconda. The browser address bar displays `docs.anaconda.com/miniconda/`. The page features a dark theme with a green Anaconda logo in the top left. A navigation sidebar on the left lists sections like 'Anaconda Documentation', 'Getting Started', 'Tools', and 'Business Solutions', with 'Miniconda' highlighted under 'Getting Started'. The main content area has a breadcrumb 'Home > Miniconda' and a large heading 'Miniconda'. Below the heading, it describes Miniconda as a free, miniature installation of Anaconda Distribution. It includes instructions on how to install more packages using the `conda install` command. There are two green buttons with white text: 'Is Miniconda free for me?' and 'Should I install Miniconda or Anaconda Distribution?'. Below these, a section titled 'Latest Miniconda installer links' provides instructions for downloading the latest installers for Python 3.12. The first step, 'Go to anaconda.com/download', is highlighted with a red rectangle. The right sidebar contains a search bar and a list of links for 'On this page'.

docs.anaconda.com/miniconda/

More Resources ▾ Download Anaconda Pricing Search Ctrl + K

Home > Miniconda

Miniconda

Miniconda is a free, miniature installation of Anaconda Distribution that includes only conda, Python, the packages they both depend on, and a small number of other useful packages.

If you need more packages, use the `conda install` command to install from thousands of packages available by default in Anaconda's public repo, or from other channels, like conda-forge or bioconda.

[Is Miniconda free for me?](#)

[Should I install Miniconda or Anaconda Distribution?](#)

Latest Miniconda installer links

To download the latest Miniconda installers for Python 3.12:

1. Go to anaconda.com/download.
2. Submit your email to register with Anaconda, if desired.
3. Click **Submit** to continue to the Download Now page.
4. Scroll down past the Anaconda Distribution installers.

On this page
Latest Miniconda installer lin
Quick command line install

파이썬 설치 - 구글에서 '미니콘다 다운로드' 검색

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Distribution

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Register to get everything you need to get started on your workstation including Cloud Notebooks, Navigator, AI Assistant, Learning and more.

- ✓ Easily search and install thousands of data science, machine learning, and AI packages
- ✓ Manage packages and environments from a desktop application or work from the command line
- ✓ Deploy across hardware and software platforms
- ✓ Distribution installation on Windows, MacOS, or Linux

Provide email to download Distribution

Email Address:

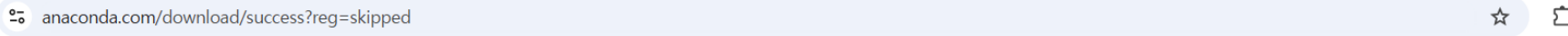
☐ Agree to receive communication from Anaconda regarding relevant content, products, and services. I understand that I can revoke this consent [here](#) at any time.

By continuing, I agree to Anaconda's [Privacy Policy](#) and [Terms of Service](#).

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Skip registration

파이썬 설치 - 구글에서 '미니콘다 다운로드' 검색



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Anaconda Distribution

Package and environment management with conda, Anaconda Navigator desktop app, and 600+ packages. Everything you need for data science.

GRAPHICAL INSTALLER

[↓ Windows 64-Bit Graphical Installer](#)

[📘 Graphical install instructions](#)

COMMAND LINE

[📄 CLI install instructions](#)

Miniconda

Lightweight installer that only includes conda, Python, and their dependencies. Install only what you need.

GRAPHICAL INSTALLER

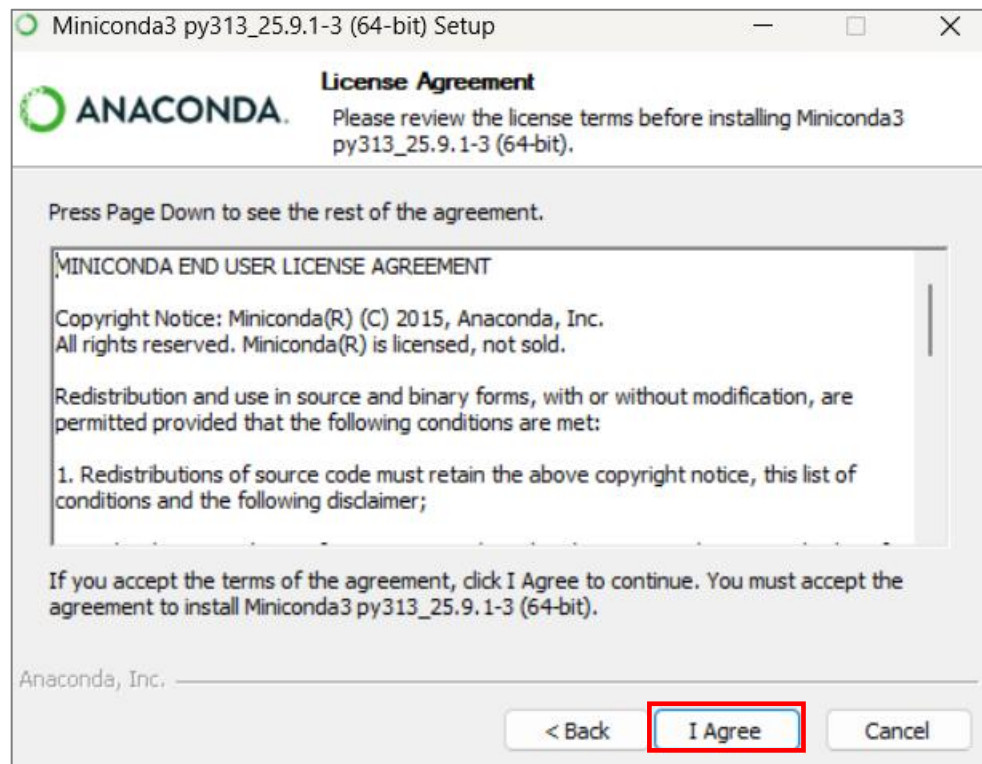
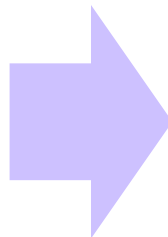
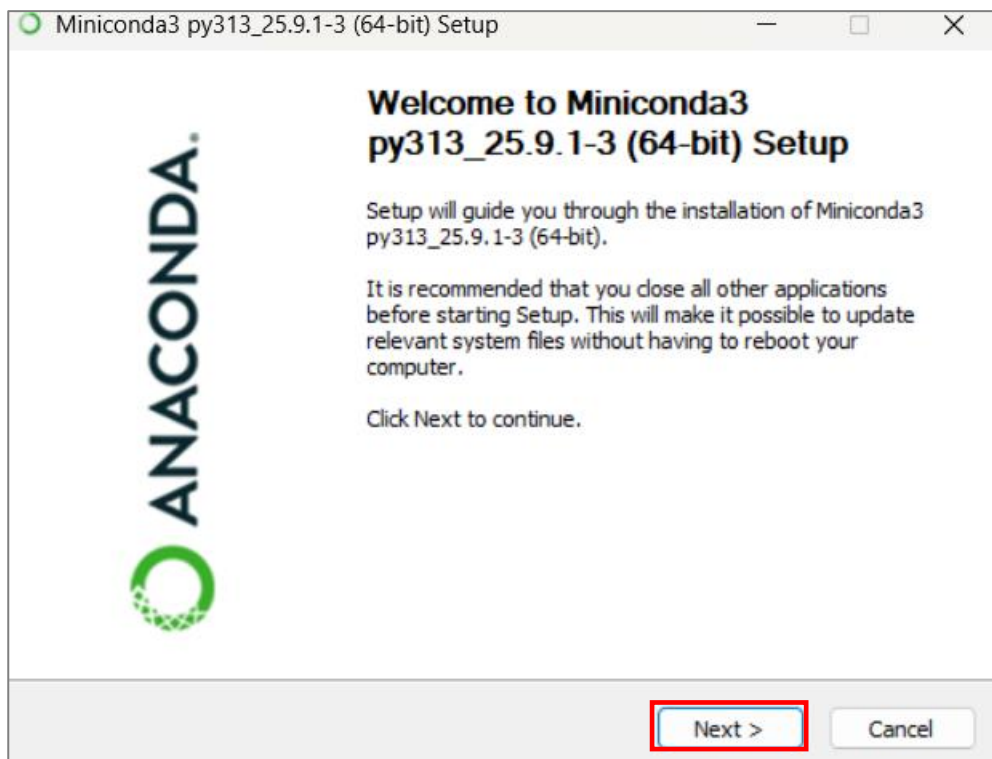
[↓ Windows 64-Bit Graphical Installer](#)

[📘 Graphical install instructions](#)

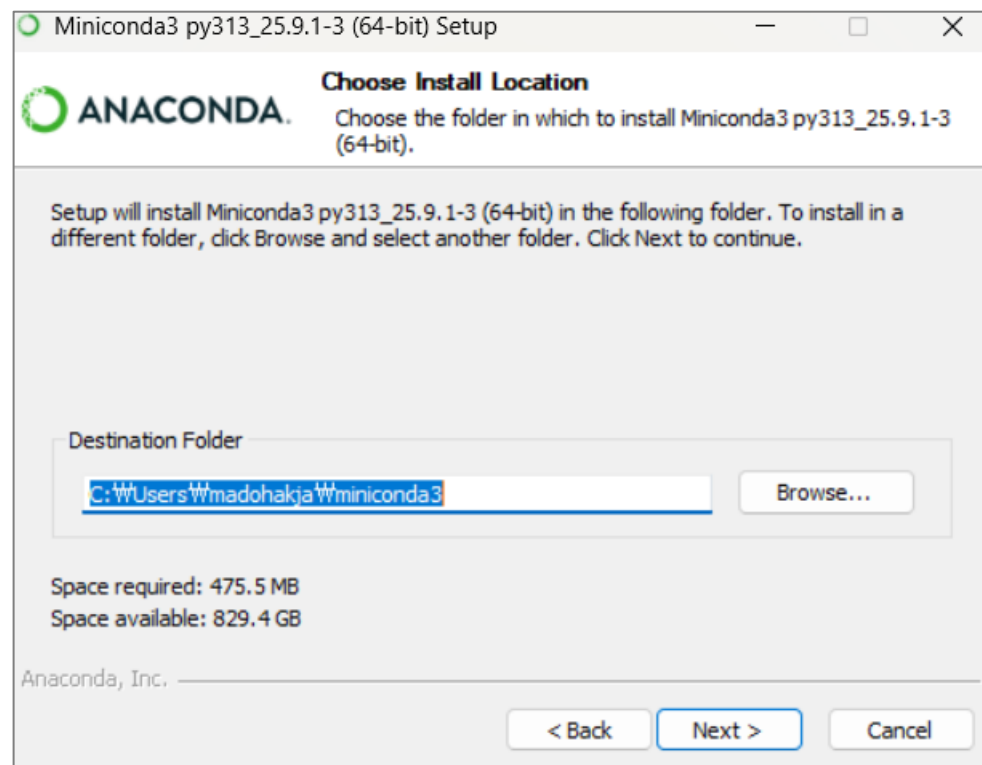
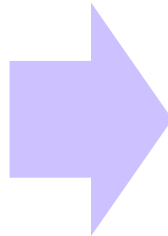
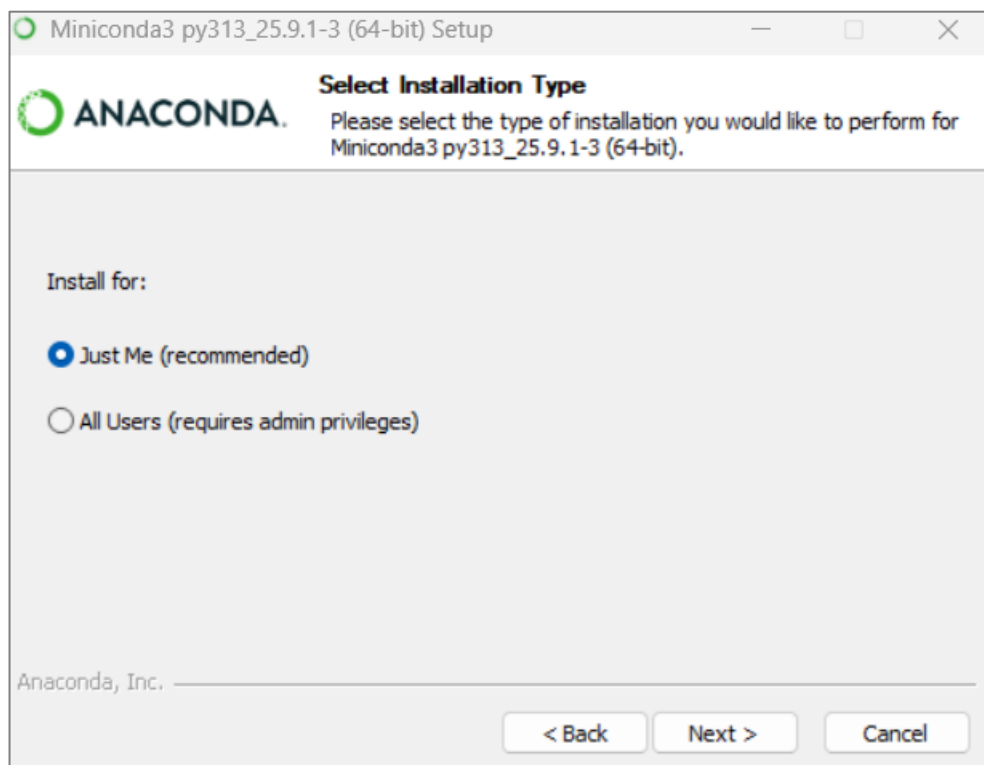
COMMAND LINE

[📄 CLI install instructions](#)

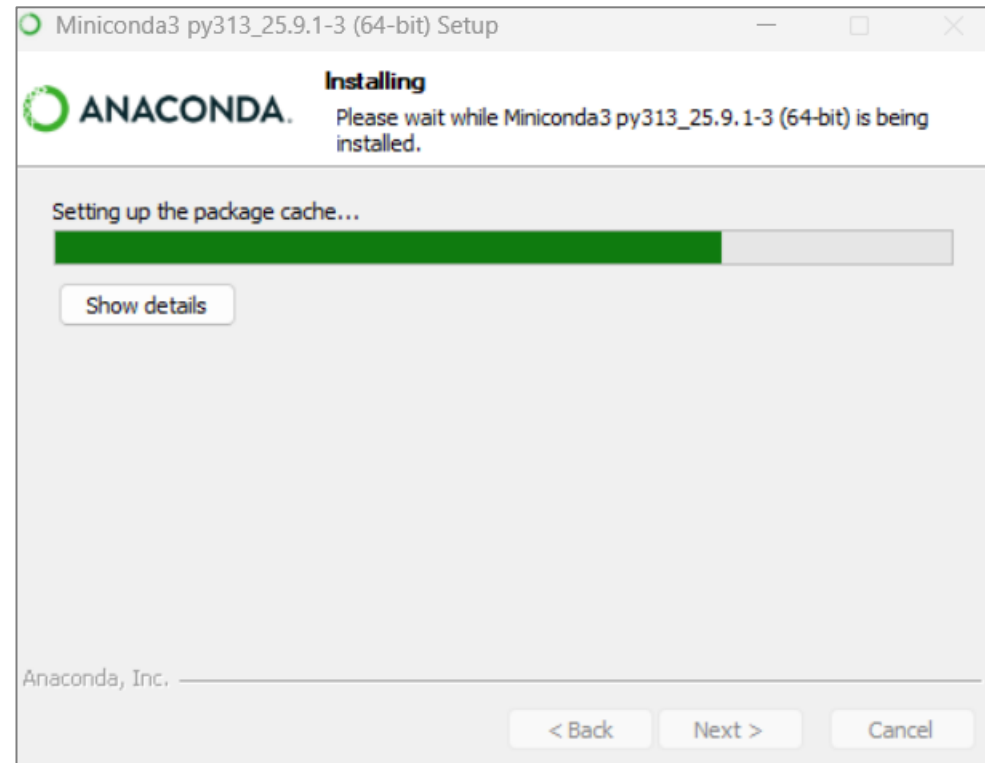
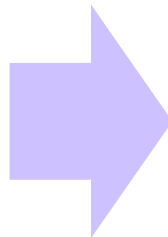
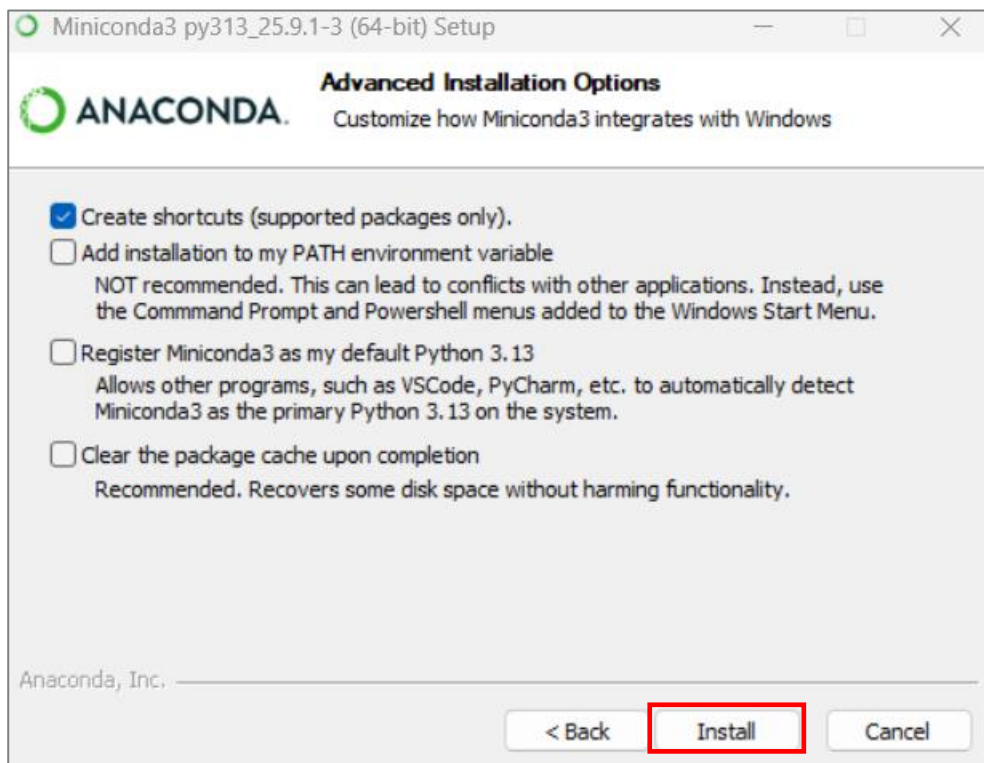
미니콘다 설치



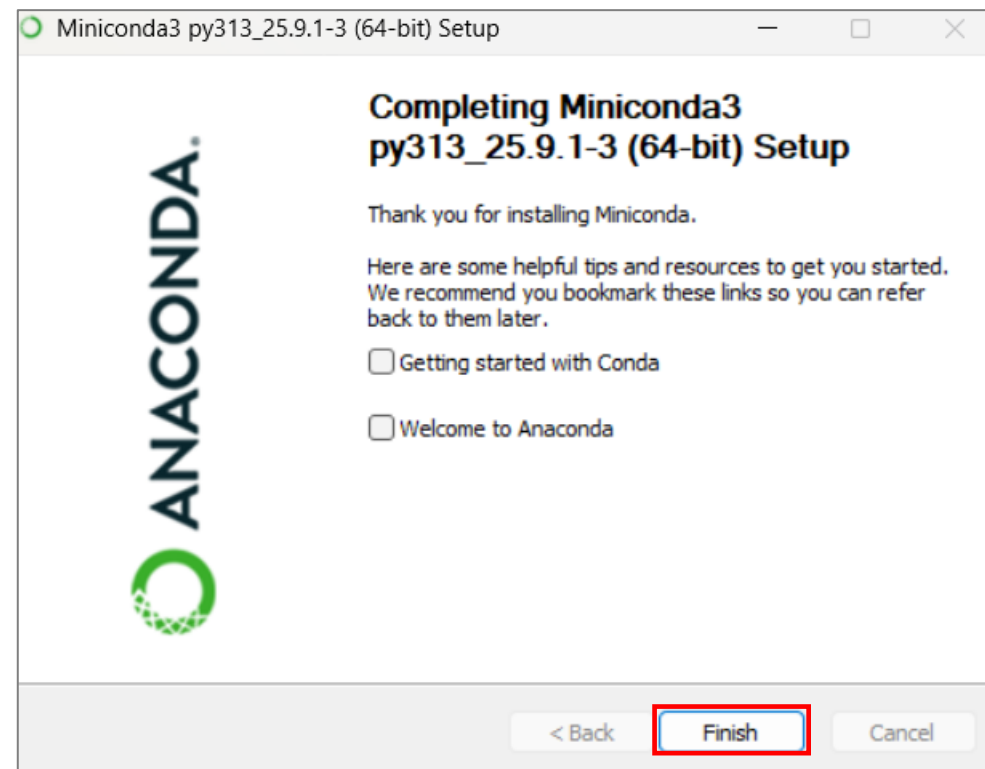
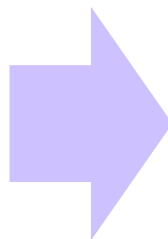
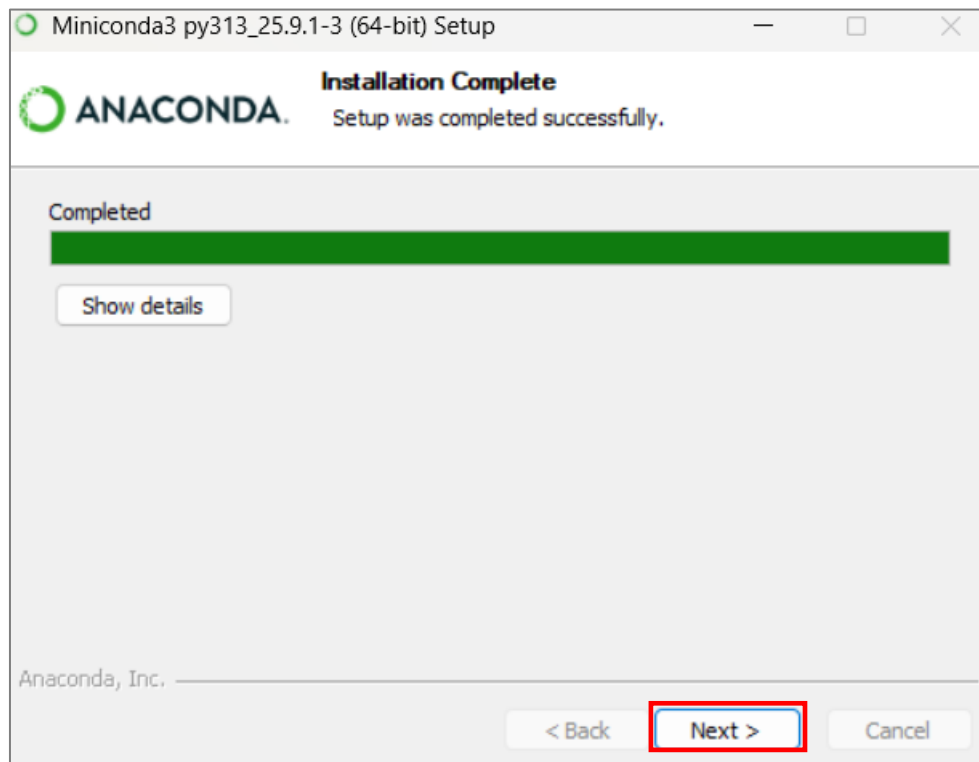
미니콘다 설치



미니콘다 설치

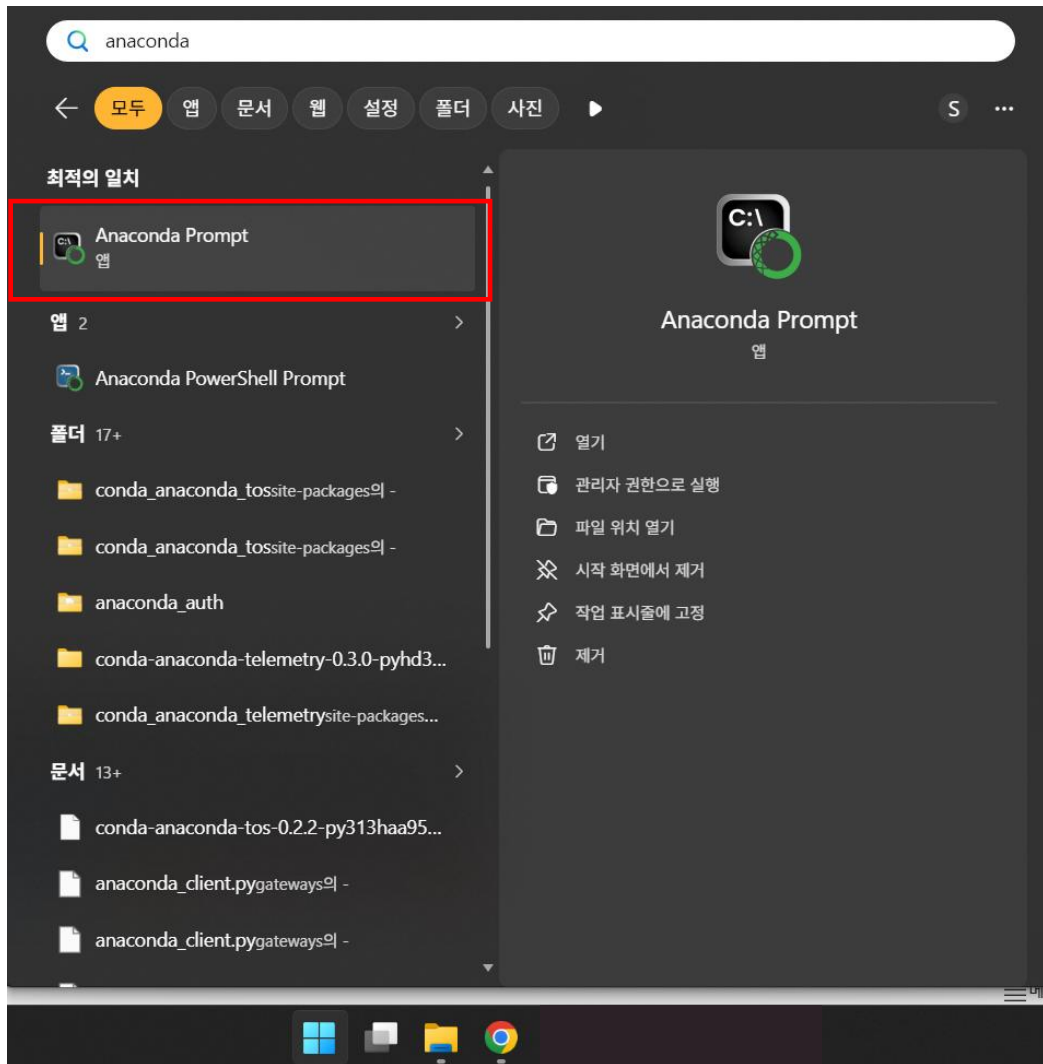


미니콘다 설치



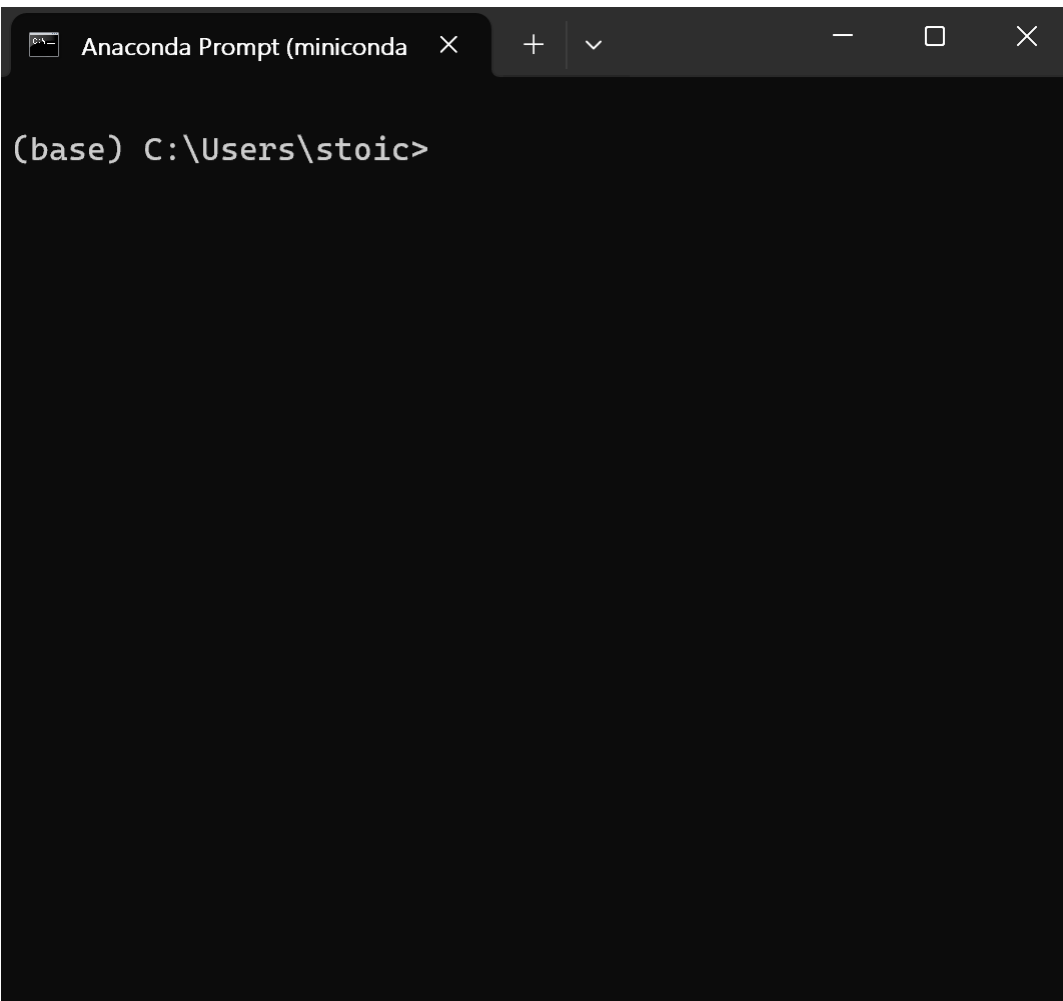
Section 파이썬 설치

미니콘다 실행



윈도우 버튼 -> anaconda 검색 -> Anaconda Prompt 실행

미니콘다 설정



가상환경 확인

```
(base) C:\Users\stoic>conda info --envs
```

```
# conda environments:
```

```
#
```

```
base * C:\ProgramData\miniconda3
```

설치 가능한 파이썬 확인

```
(base) C:\Users\stoic>conda search python
```

```
Loading channels: done
```

# Name	Version	Build	Channel
python	2.7.13	h1b6d89f_16	pkgs/main
python	2.7.13	h9912b81_15	pkgs/main
python	2.7.13	hb034564_12	pkgs/main
.			
.			
.			
python	3.11.8	he1021f5_0	pkgs/main
python	3.11.9	he1021f5_0	pkgs/main
python	3.12.0	h1d929f7_0	pkgs/main
python	3.12.1	h1d929f7_0	pkgs/main
python	3.12.2	h1d929f7_0	pkgs/main
python	3.12.3	h1d929f7_0	pkgs/main

가상환경 설치

```
(base) C:\Users\stoic>conda create --name py3_11_9 python=3.11.9
```

```
Retrieving notices: ...working... done
```

```
Channels:
```

```
- defaults
```

```
Platform: win-64
```

```
Collecting package metadata (repodata.json): done
```

```
Solving environment: done
```

```
.
```

```
.
```

```
.
```

```
Executing transaction: done
```

```
#
```

```
# To activate this environment, use
```

```
#
```

```
#     $ conda activate py3_11_9
```

```
#
```

```
# To deactivate an active environment, use
```

```
#
```

```
#     $ conda deactivate
```

가상환경 확인 및 실행

```
(base) C:\Users\stoic>conda info --envs
# conda environments:
#
base                * C:\ProgramData\miniconda3
py3_11_9            C:\Users\stoic\.conda\envs\py3_11_9
```

```
(base) C:\Users\stoic>conda activate py3_11_9   가상환경 실행
```

```
(py3_11_9) C:\Users\stoic>conda deactivate     가상환경 해제
```

```
(base) C:\Users\stoic>
```

라이브러리 설치

```
(base) C:\Users\stoic>conda activate py3_11_9
```

```
(py3_11_9) C:\Users\stoic>pip install numpy
```

```
(py3_11_9) C:\Users\stoic>pip install pyarrow
```

```
(py3_11_9) C:\Users\stoic>pip install pandas
```

```
(py3_11_9) C:\Users\stoic>pip install scipy
```

```
(py3_11_9) C:\Users\stoic>pip install matplotlib
```

```
(py3_11_9) C:\Users\stoic>pip install seaborn
```

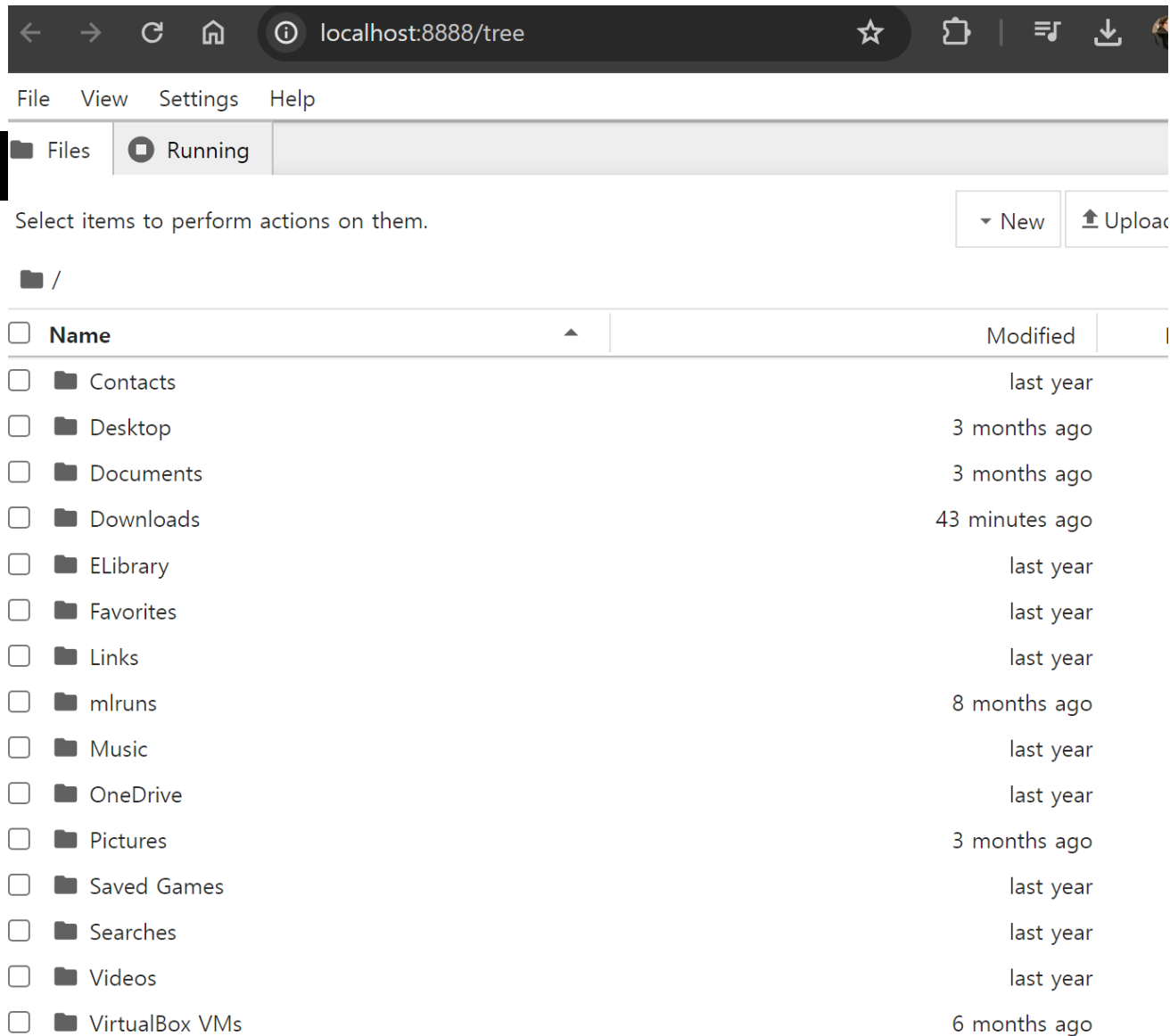
```
(py3_11_9) C:\Users\stoic>pip install scikit-learn
```

```
(py3_11_9) C:\Users\stoic>pip install prophet
```

```
(py3_11_9) C:\Users\stoic>pip install jupyter
```

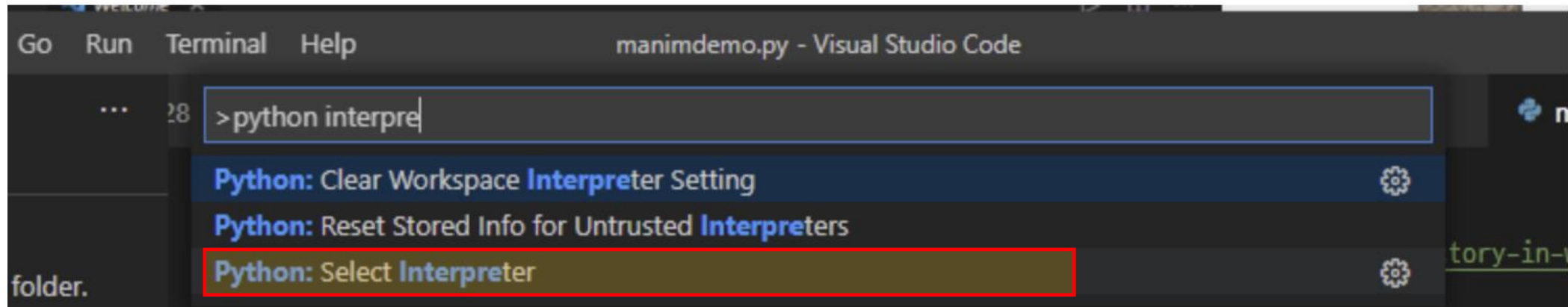
주피터 노트북 실행

```
(py3_11_9) C:\Users\stoic>jupyter notebook
```

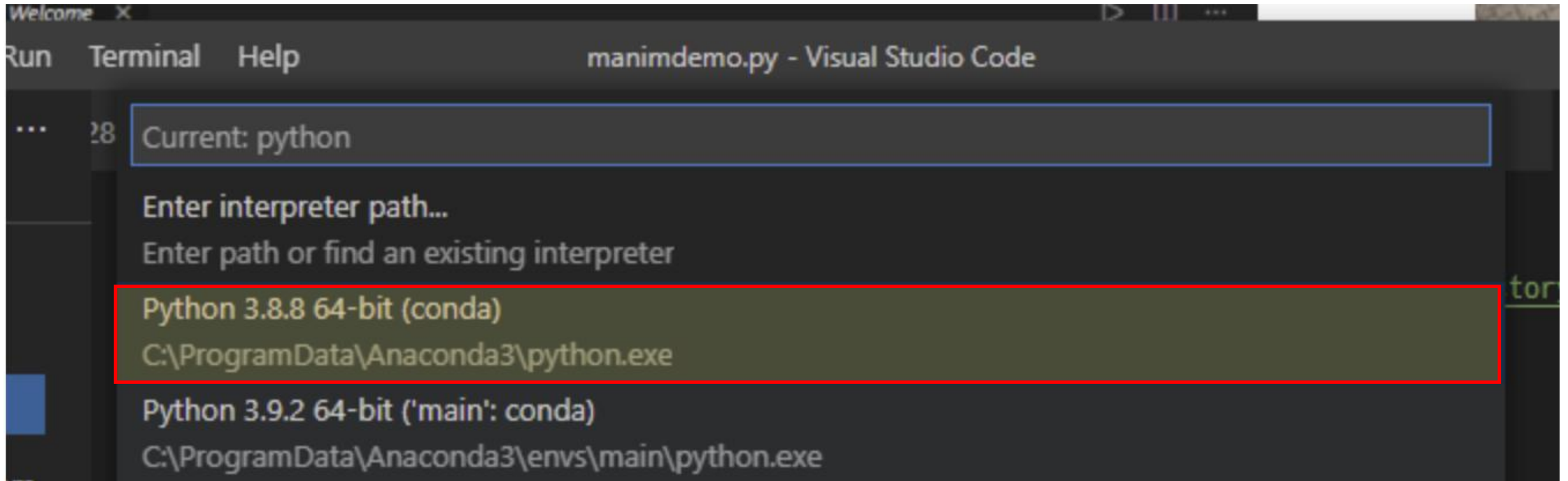


파이썬 설정 – vscode와 아나콘다 연동

control + shift + p ► python interpreter 검색

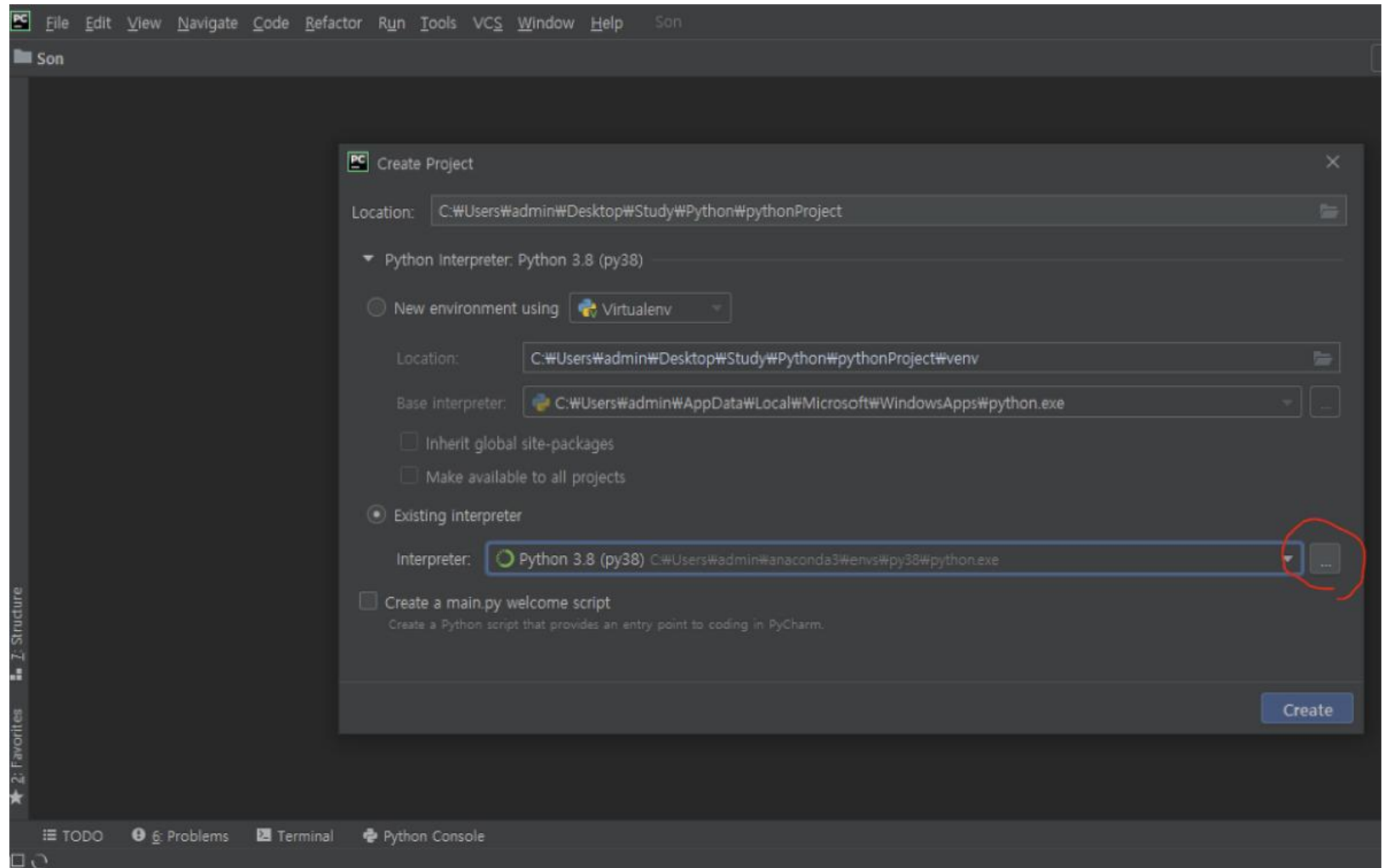


파이썬 설정 – vscode와 아나콘다 연동



파이썬 설정 - 파이참과 아나콘다 연동

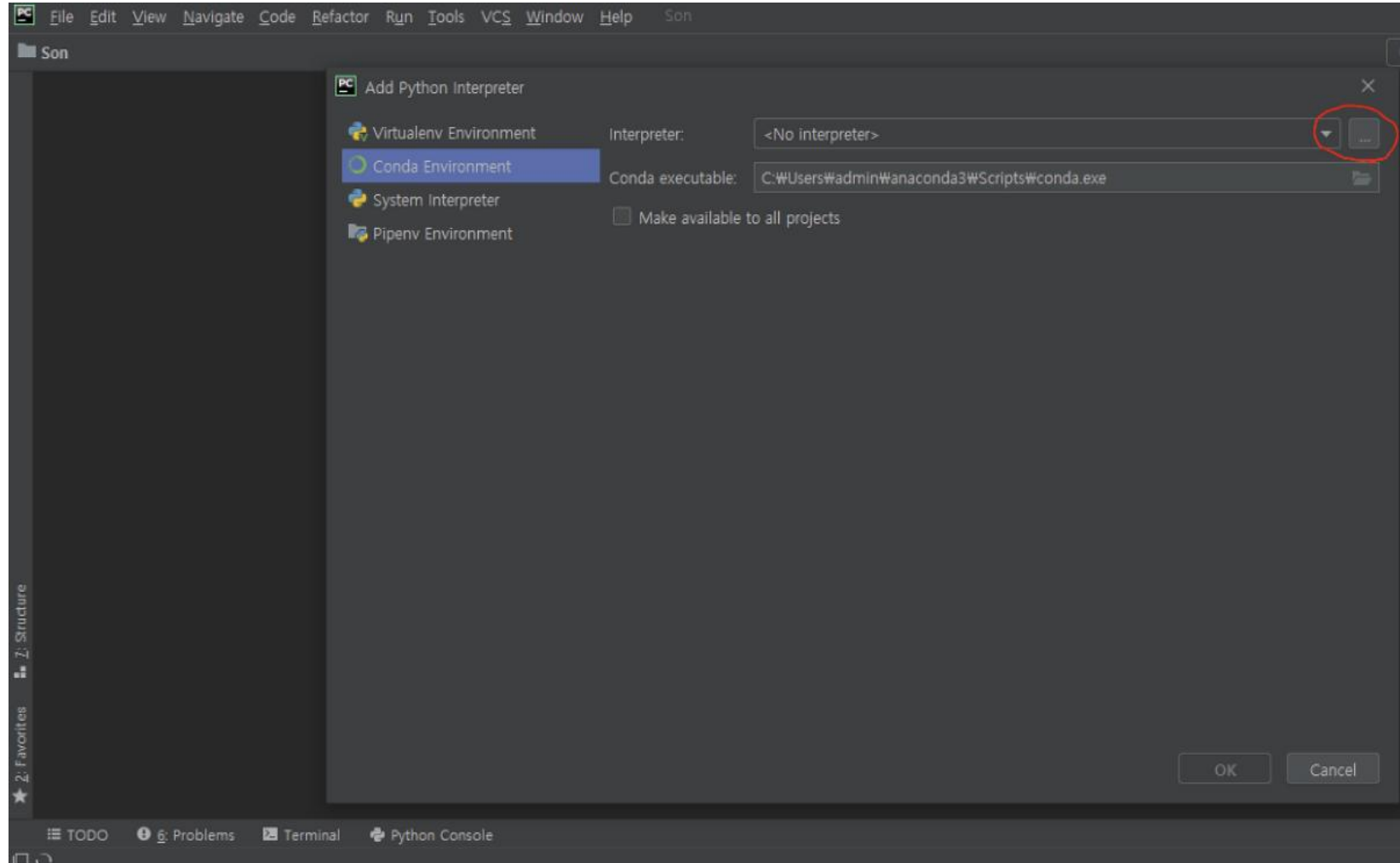
1. create new project
2. location 지정
3. Existing interpreter 설정 클릭



파이썬 설정 - 파이참과 아나콘다 연동

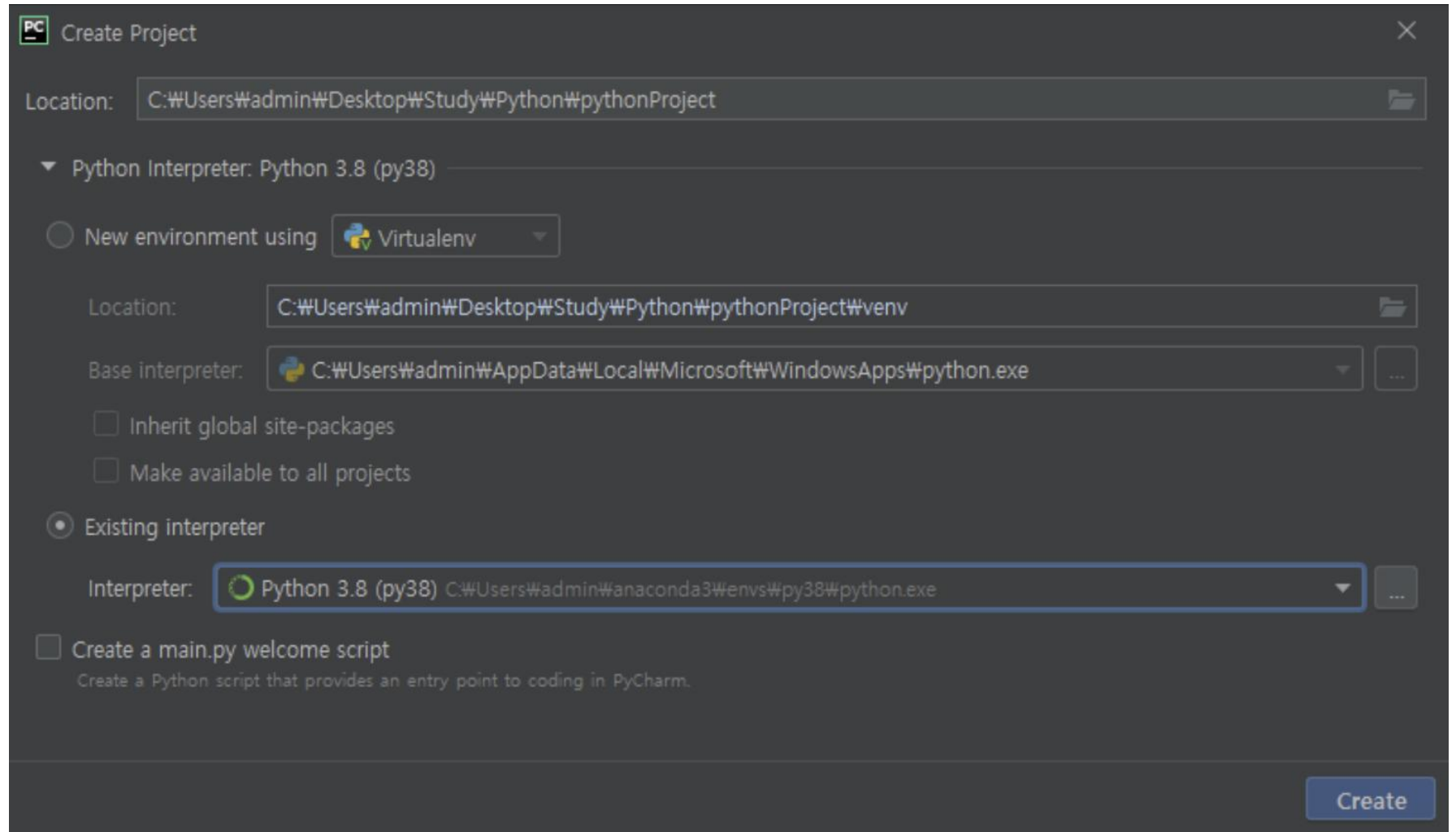
Conda Environment

에서 인터프리터 설정

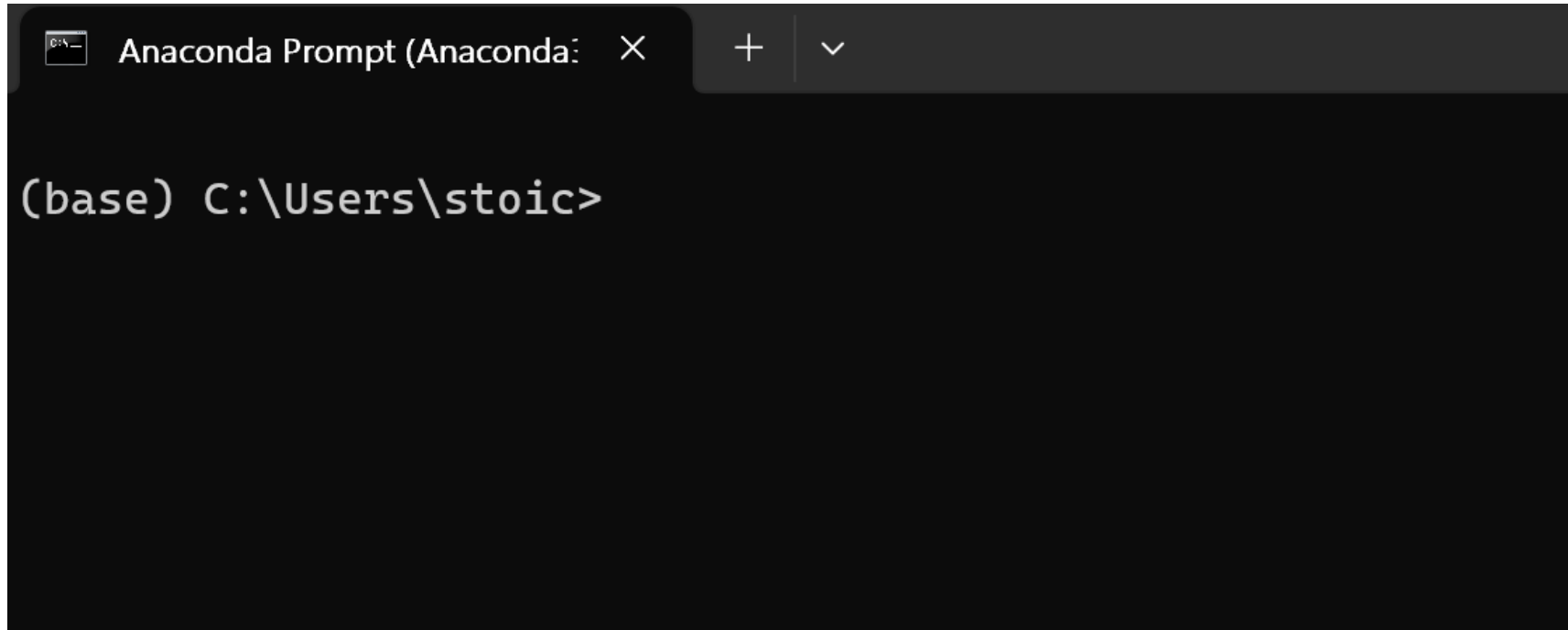


파이썬 설정 - 파이참과 아나콘다 연동

프로젝트 생성



파이썬 설정 - 프록시 설정



```
Anaconda Prompt (Anaconda:  ×  +  ∨  
  
(base) C:\Users\stoic>
```

파이썬 설정 - 프록시 설정

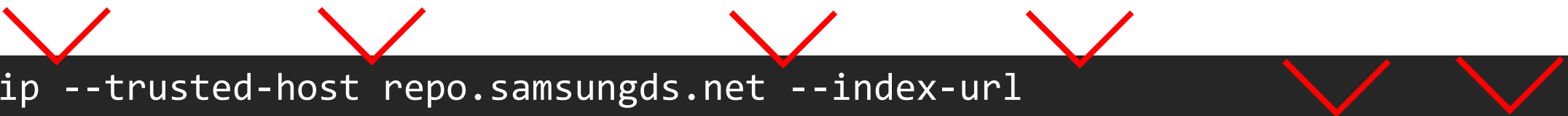


```
pip --trusted-host pypi.org --trusted-host files.pythonhosted.org install seaborn
```



```
conda config --set ssl_verify false
```

파이썬 설정 - 프록시 설정



```
pip --trusted-host repo.samsungds.net --index-url  
http://repo.samsungds.net:8081/artifactory/api/pypi/pypi/simple install  
numpy pandas matplotlib seaborn
```



```
conda config --set ssl_verify false
```

파이썬 설정 - 프록시 설정

```
[global]  
trusted-host= repo.samsungds.net  
index-url= http://repo.samsungds.net:8081/artifactory/api/pypi/pypi/simple
```

파이썬 설정 - 프록시 설정

C:\Users\<username>\.condarc

파일 생성

channels:

- r
- defaults

ssl_verify: False

show_channel_urls: True

allow_other_channels: True

proxy_servers:

https: https://pypi.org

https: <https://files.pythonhosted.org/>

https: <https://repo.samsungds.net>

AI/BigData 인텐시브 과정

Section 2. 환경 설정

Section 2-3. 주피터 노트북 기본 사용법

주피터노트북 실행

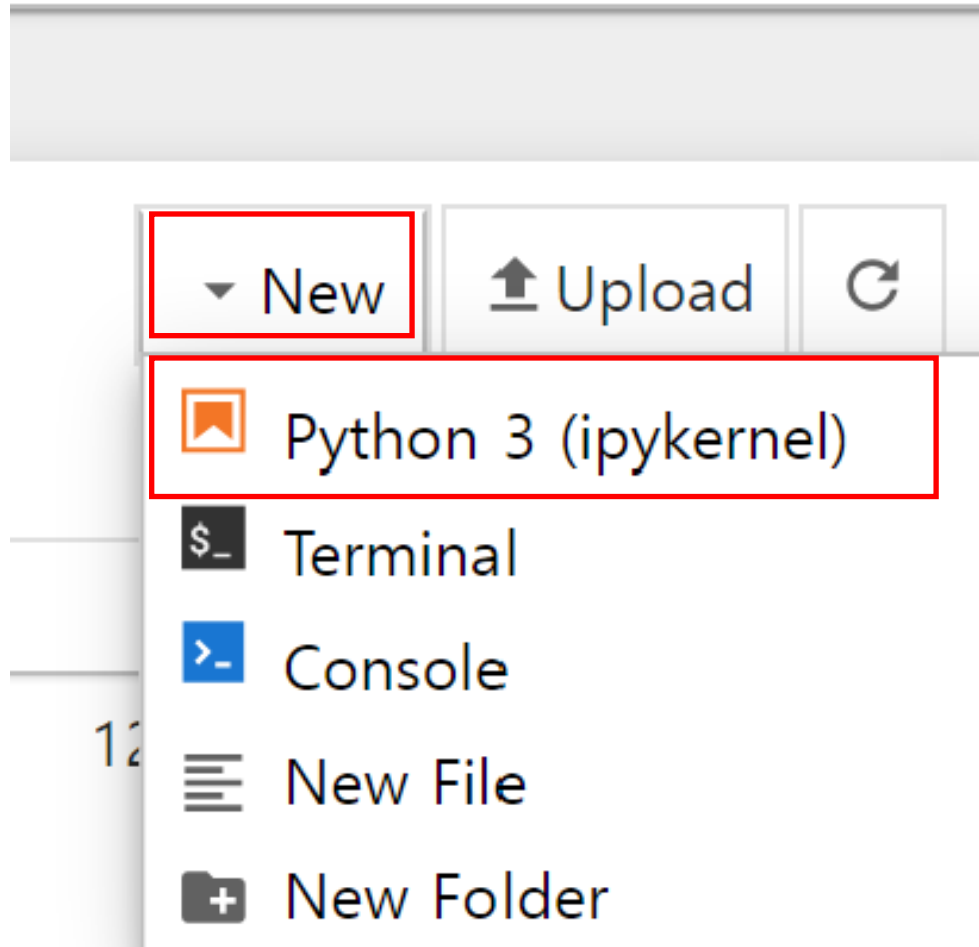


새 폴더



practice

주피터노트북 실행



주피터노트북 실행

 jupyter Untitled2 Last Checkpoint: 21 seco

File Edit View Run Kernel Settings Help

         Code ▼

또는 shift + enter

```
[1]: 1+1
```

```
[1]: 2
```

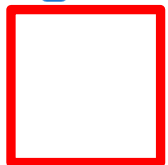
주피터 노트북 소개 - 셀 위로 삽입

 jupyter Untitled2 Last Checkpoint: 1 minute ago

File Edit View Run Kernel Settings Help

         Code ▾

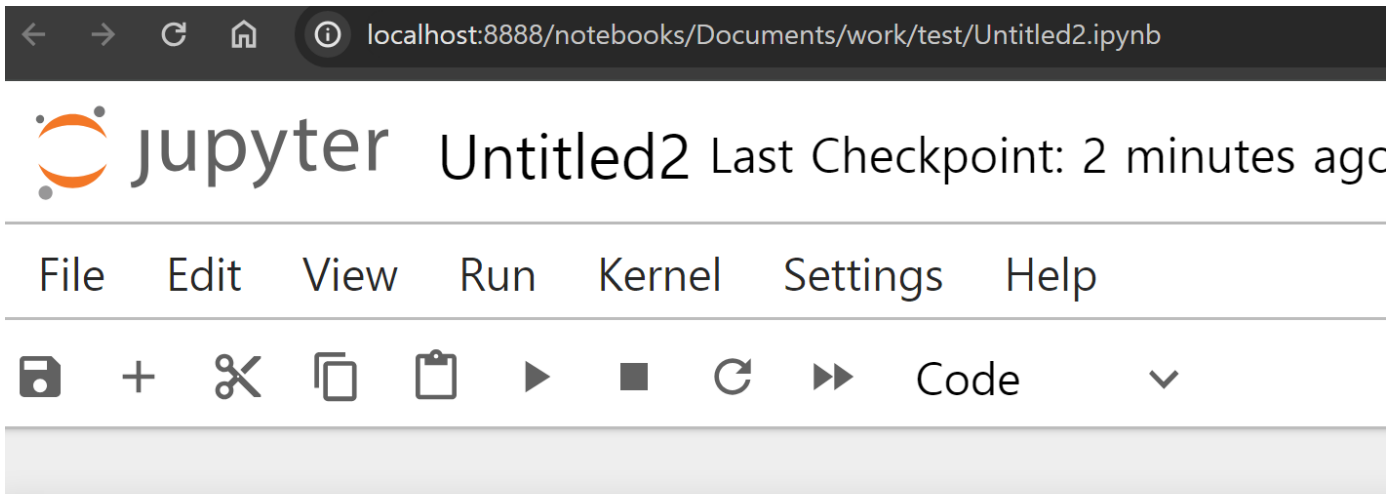
 []:



[1]: 1+1

a키 입력 [1]: 2

주피터 노트북 소개 - 셀 아래로 삽입



[]:

[1]: 1+1

[1]: 2

[]:

또는 b

주피터 노트북 소개 - 셀 삭제



```
[1]: 1+1
```

```
[1]: 2
```

dd



```
[ ]:
```

주피터 노트북 소개 - 실행 취소

 jupyter Untitled2 Last Checkpoint: 5 minutes ago

File Edit View Run Kernel Settings Help

 +        Code ▾

```
[ ]:   
[ ]: 
```

 jupyter Untitled2 Last Checkpc

File Edit View Run Kernel Settings

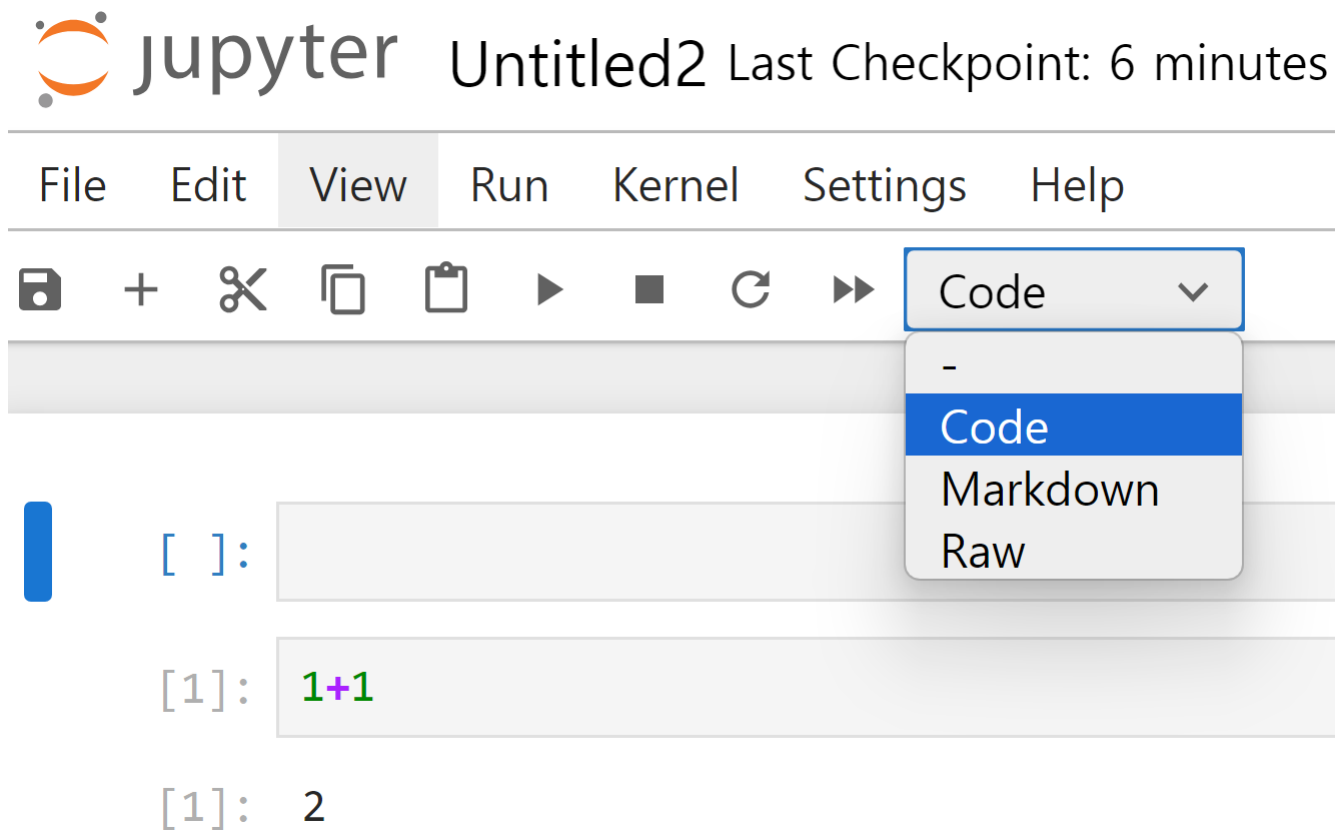
 +        Coc

```
[1]: 1+1
```

```
[1]: 2
```

```
[ ]: 
```

주피터 노트북 소개 - 마크다운 모드로 변경



주피터 노트북 소개 - 초기화



Jupyter

Untitled2 Last Checkpoint: 7 minutes ago

File Edit View Run Kernel Settings Help



Interrupt Kernel

Restart Kernel...

Restart Kernel and Clear Outputs of All Cells...

Restart Kernel and Run up to Selected Cell...

Restart Kernel and Run All Cells...

Restart Kernel and Debug...

Reconnect to Kernel

Shut Down Kernel

Shut Down All Kernels...

Change Kernel...

Restart Kernel?

Do you want to restart the kernel of Untitled2.ipynb? All variables will be lost.

Cancel

Restart

주피터 노트북 소개 - 주석

In [1]:

```
1 # 파이썬 기초  
2 1+1 # 더하기를 계산하는 코드
```

Out[1]: 2

AI/BigData 인텐시브 과정

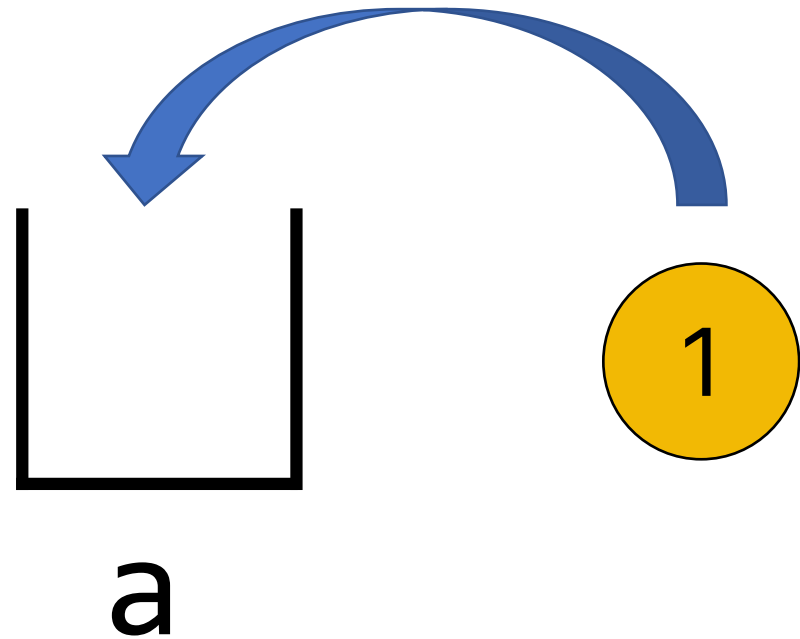
Section 3. 파이썬 기초

Section 3-1. 파이썬 기본 문법

숫자형 변수

```
In [9]: 1 a = 1
In [10]: 1 print(a)
          1
In [11]: 1 a
Out[11]: 1

In [20]: 1 type(a)
Out[20]: int
```



숫자형 변수

In [9]:

1	a = 1
---	-------



In []:

1	
---	--



In [2]:

1	a
---	---

Out[2]: 1

셀을 삭제해도 변수는 사라지지 않습니다.

숫자형 변수

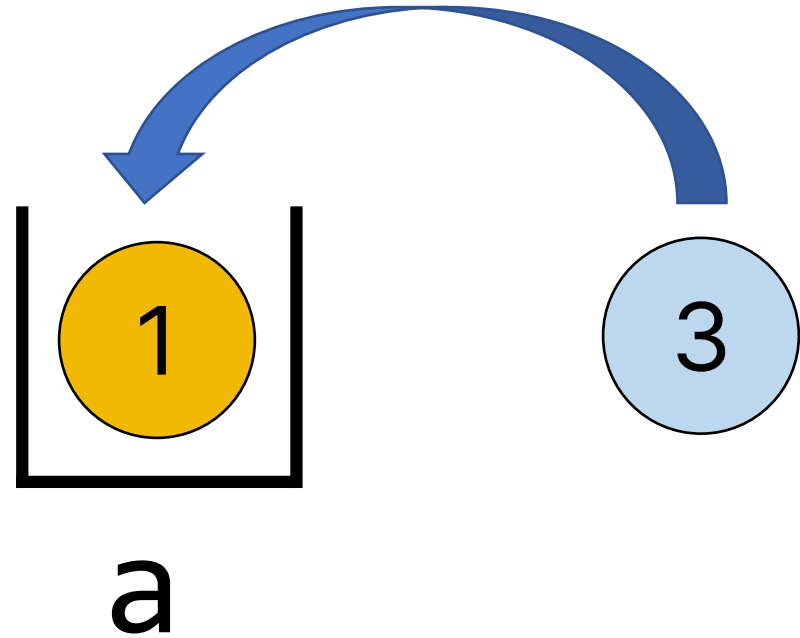
```
In [12]:
```

```
1 a = 3
```

```
In [13]:
```

```
1 print(a)
```

3



숫자형 변수

In [16]:

```
1 c = a + b
2 print(a)
3 print(b)
4 print(c)
```

```
3
2
5
```

숫자형 변수로 수학 연산 가능

숫자형 변수

$a - b = ?$

$a * b = ?$

$a / b = ?$

문자형 변수

```
In [17]: 1 string1 = "hello"  
        2 print(string1)
```

hello

```
In [19]: 1 type(string1)
```

Out[19]: str

```
In [21]: 1 string2 = ", world"
```

```
In [24]: 1 string3 = string1 + string2
```

```
In [25]: 1 print(string3)
```

hello, world

문자형 변수

In [27]:

1	<code>x = 1</code>
2	<code>type(x)</code>

Out[27]: `int`

In [28]:

1	<code>y = "1"</code>
2	<code>type(y)</code>

Out[28]: `str`

1 ≠ "1"

리스트(list)?

대괄호[] 사용

In [29]:

```
1 a = [1, 2, 3, 4, 5]
2 print(a)
```

[1, 2, 3, 4, 5]

여러 데이터를 한번에 저장 가능

리스트(list)?

```
In [29]:
```

1	a = [1, 2, 3, 4, 5]
2	print(a)

[1, 2, 3, 4, 5]

0 1 2 3 4

```
In [30]:
```

1	a[0]
---	------

```
Out[30]: 1
```

```
In [31]:
```

1	a[1]
---	------

```
Out[31]: 2
```

```
In [32]:
```

1	a[2]
---	------

```
Out[32]: 3
```

```
In [33]:
```

1	a[3]
---	------

```
Out[33]: 4
```

```
In [34]:
```

1	a[4]
---	------

```
Out[34]: 5
```

리스트(list) - append

```
In [29]: 1 a = [1, 2, 3, 4, 5]
          2 print(a)
[1, 2, 3, 4, 5]
```

```
In [35]: 1 a.append(6)
```

```
In [36]: 1 print(a)
[1, 2, 3, 4, 5, 6]
```

리스트(list) 속 리스트

In [55]:

```
1 b = [ 1, 2, [3,4] ]  
2 print(b)
```

[1, 2, [3, 4]]

In [56]:

```
1 b[0]
```

Out[56]: 1

In [57]:

```
1 b[1]
```

Out[57]: 2

In [58]:

```
1 b[2]
```

Out[58]: [3, 4]

In [59]:

```
1 b[2][0]
```

Out[59]: 3

In [60]:

```
1 b[2][1]
```

Out[60]: 4

조건문 - if문

In [50]:

```
1 a = 3
2 if a > 2:
3     print("a는 2보다 크다")
```

a는 2보다 크다

In [51]:

```
1 a = 1
2 if a > 2:
3     print("a는 2보다 크다")
```

파이썬은 **들여쓰기**가 중요

In [160]:

```
1 a = 1
2 if a > 2:
3     print("a는 2보다 크다")
4 else:
5     print("a는 2보다 작거나 같다")
```

a는 2보다 작거나 같다

In [161]:

```
1 a = 1.5
2 if a > 2:
3     print("a는 2보다 크다")
4 elif a > 1:
5     print("a는 1보다 크고 2보다 작거나 같다")
6 else:
7     print("a는 1보다 작거나 같다")
```

a는 1보다 크고 2보다 작거나 같다

반복문 - for문

In [61]:

```
1 for i in [0,1,2,3]:  
2     print(i)
```

0
1
2
3

In [62]:

```
1 for i in range(0,4):  
2     print(i)
```

0
1
2
3

반복문 - for문

In [63]:

```
1 for i in ["a", "b", "c"]:  
2     print(i)
```

a
b
c

In [64]:

```
1 item_list = ["a", "b", "c"]  
2 for i in item_list:  
3     print(i)
```

a
b
c

AI/BigData 인텐시브 과정

Section 3. 파이썬 기초

Section 3-2. pandas 라이브러리 기초

pandas 라이브러리

```
In [32]: 1 import pandas as pd
```

- `import pandas :` pandas 라이브러리를 불러온다.
- `as pd :` pd라고 짧게 줄여서 부르겠다.

Section

pandas 라이브러리 기초

```
In [4]: 1 import pandas as pd
```

```
In [6]: 1 df = pd.read_csv("./performance.csv", encoding='cp949')
```

```
In [7]: 1 df
```

Out[7]:

		팀	학력	사내교육	성과	능력	태도
	GENDER	TEAM	EDUCATION	TRAINING	ACHIEVEMENT	ABILITY	ATTITUDE
0	female	B	bachelor's degree	none	72	72	74
1	female	C	some college	completed	69	90	88
2	female	B	master's degree	none	90	95	93
3	male	A	associate's degree	none	47	57	44
4	male	C	some college	none	76	78	75
...	...	...	...	...	...	...	...
995	female	E	master's degree	completed	88	99	95
996	male	C	high school	none	62	55	55
997	female	C	high school	completed	59	71	65
998	female	D	some college	completed	68	78	77
999	female	D	some college	none	77	86	86

1000 rows × 7 columns

열 이름, 변수 명, 필드 명

The diagram illustrates a 2D array structure. It consists of a grid of 15 rows and 8 columns. The first row is shaded gray and is labeled "행(row)" on the left. The first column is shaded gray and is labeled "열(column)" at the bottom. A red box highlights the top-left cell, which is the intersection of the first row and the first column. Below the grid, the text "레코드, 관측치" (Record, Observation) is written, indicating that each cell in the grid represents a record or observation.

개발자에게 유리한 언어

한글

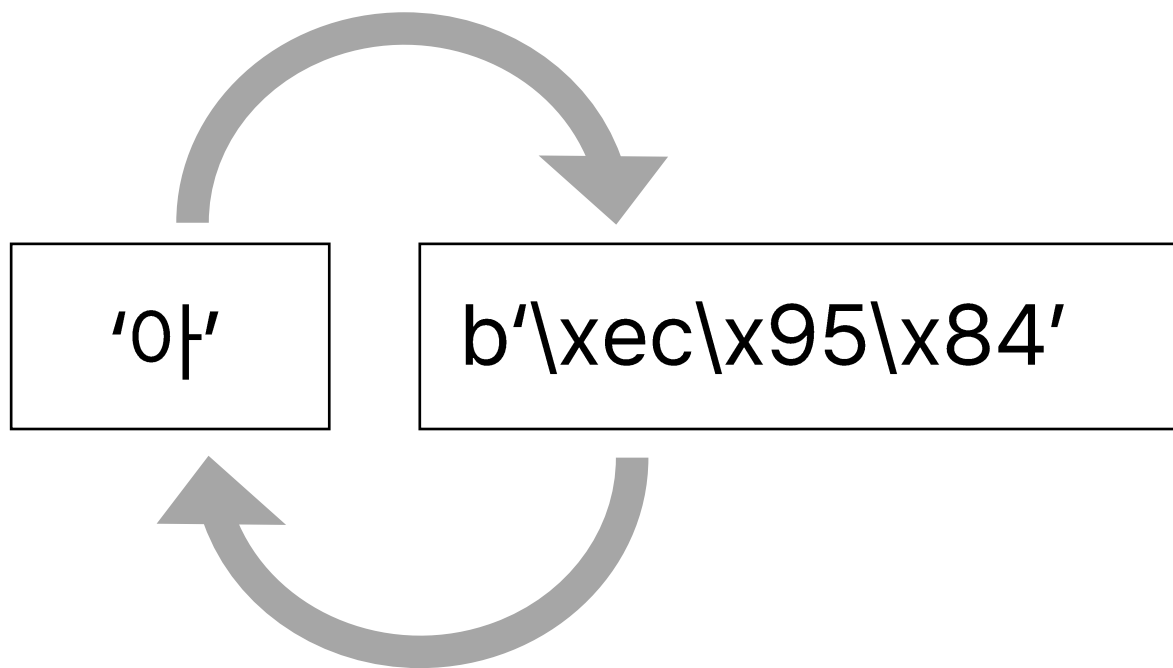
VS

영어

- 엑셀에서 한글 사용
- 컴퓨터 사용자 이름 한글로 설정
- ...

인코딩(encoding)

인코딩(encoding): 컴퓨터가 읽을 수 있는 형태로 변환



디코딩(decoding): 인간이 읽을 수 있는 형태로 변환

상단/하단 데이터만 추출

```
In [8]: 1 df.head(5)
```

Out[8]:

	GENDER	TEAM	EDUCATION	TRAINING	ACHIEVEMENT	ABILITY	ATTITUDE
0	female	B	bachelor's degree	none	72	72	74
1	female	C	some college	completed	69	90	88
2	female	B	master's degree	none	90	95	93
3	male	A	associate's degree	none	47	57	44
4	male	C	some college	none	76	78	75

```
In [88]: 1 df.tail(3)
```

Out[88]:

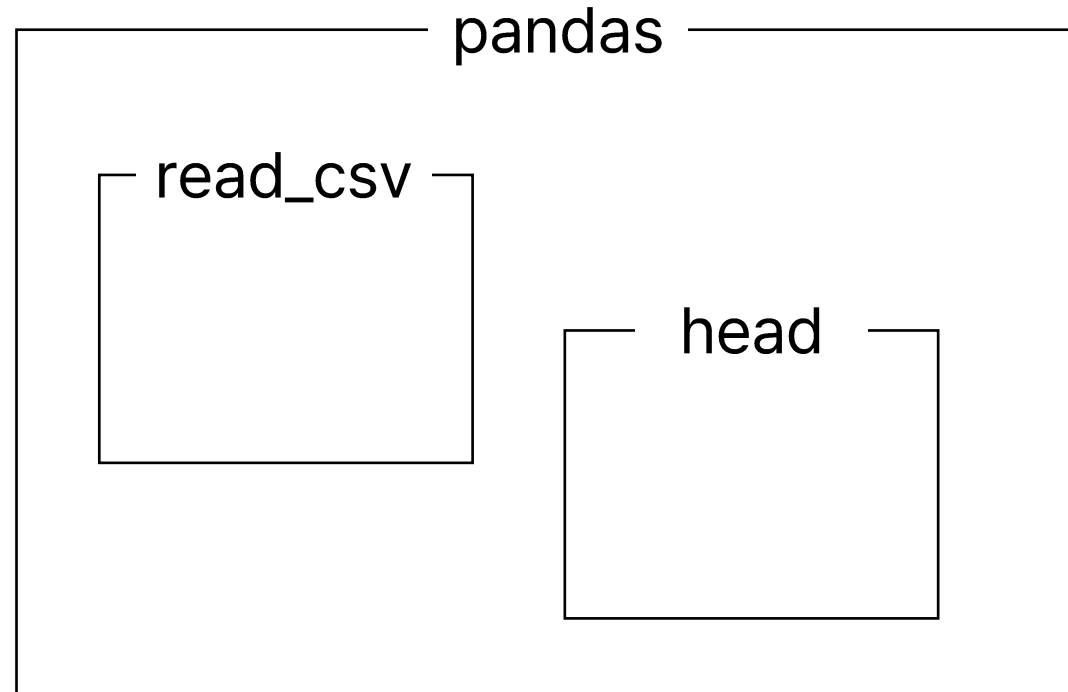
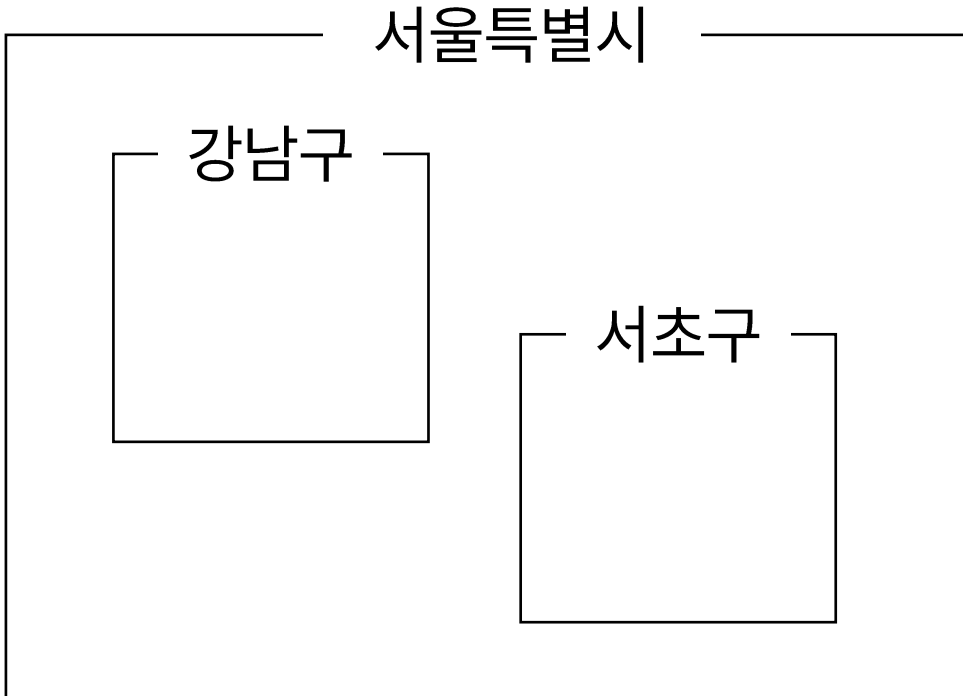
	GENDER	TEAM	EDUCATION	TRAINING	ACHIEVEMENT	ABILITY	ATTITUDE
997	female	C	high school	completed	59	71	65
998	female	D	some college	completed	68	78	77
999	female	D	some college	none	77	86	86

점(.)은 뭔가요?

```
In [4]: ▶ 1 import pandas as pd
```

```
In [6]: ▶ 1 df = pd.read_csv("../performance.csv")
```

서울특별시 강남구 ~~~



열(column)/변수 이름 추출

In [89]: ▶

```
1 df.columns
```

```
Out[89]: Index(['GENDER', 'TEAM', 'EDUCATION', 'TRAINING', 'ACHIEVEMENT', 'ABILITY',  
               'ATTITUDE'],  
              dtype='object')
```

데이터 크기 확인

```
In [90]: ▶ 1 df.shape
```

```
Out[90]: (1000, 7)
```

데이터 크기 확인

```
In [22]: 1 df['ABILITY']
```

```
Out[22]: 0      72
          1      90
          2      95
          3      57
          4      78
          ..
          995     99
          996     55
          997     71
          998     78
          999     86
          Name: ABILITY, Length: 1000, dtype: int64
```

유니크 값 확인

```
In [91]: ▶ 1 df['TEAM'].unique()
```

```
Out[91]: array(['B', 'C', 'A', 'D', 'E'], dtype=object)
```

새로운 열/변수 추가

```
In [92]: 1 df['TOTAL_SCORE'] = df['ACHIEVEMENT'] + df['ABILITY'] + df['ATTITUDE']
```

```
In [93]: 1 df
```

Out[93]:

	GENDER	TEAM	EDUCATION	TRAINING	ACHIEVEMENT	ABILITY	ATTITUDE	TOTAL_SCORE
0	female	B	bachelor's degree	none	72	72	74	218
1	female	C	some college	completed	69	90	88	247
2	female	B	master's degree	none	90	95	93	278
3	male	A	associate's degree	none	47	57	44	148
4	male	C	some college	none	76	78	75	229
...	...	...	...	...	...	...	...	...
995	female	E	master's degree	completed	88	99	95	282
996	male	C	high school	none	62	55	55	172
997	female	C	high school	completed	59	71	65	195
998	female	D	some college	completed	68	78	77	223
999	female	D	some college	none	77	86	86	249

1000 rows × 8 columns

기존 열/변수 삭제

```
In [99]: 1 df = df.drop(['TOTAL_SCORE'], axis='columns')  
        2 df
```

Out[99]:

	GENDER	TEAM	EDUCATION	TRAINING	ACHIEVEMENT	ABILITY	ATTITUDE
0	female	B	bachelor's degree	none	72	72	74
1	female	C	some college	completed	69	90	88
2	female	B	master's degree	none	90	95	93
3	male	A	associate's degree	none	47	57	44
4	male	C	some college	none	76	78	75
...	...	...	...	...	...	...	...
995	female	E	master's degree	completed	88	99	95
996	male	C	high school	none	62	55	55
997	female	C	high school	completed	59	71	65
998	female	D	some college	completed	68	78	77
999	female	D	some college	none	77	86	86

1000 rows × 7 columns

기존 열/변수 추출

In [22]: `1 df['ABILITY']`

Out[22]:

0	72
1	90
2	95
3	57
4	78
...	...
995	99
996	55
997	71
998	78
999	86

Name: ABILITY, Length: 1000, dtype: int64

Series

In [23]: `1 df[['ABILITY']]`

Out[23]:

	ABILITY
0	72
1	90
2	95
3	57
4	78
...	...
995	99
996	55
997	71
998	78
999	86

1000 rows × 1 columns

DataFrame

In [24]: `1 df[['ACHIEVEMENT', 'ABILITY']]`

Out[24]:

	ACHIEVEMENT	ABILITY
0	72	72
1	69	90
2	90	95
3	47	57
4	76	78
...	...	...
995	88	99
996	62	55
997	59	71
998	68	78
999	77	86

1000 rows × 2 columns

DataFrame

감사합니다.

Q & A